Multi-Source Candidate Data Transformer

Design One-Pager - Eightfold Engineering Intern Assignment

1. The Problem

Eightfold ingests candidate information from structured sources (Recruiter CSV, ATS JSON, GitHub API) and unstructured sources (Resumes, Recruiter Notes). Downstream systems require one clean, canonical profile per candidate with normalized formats, deduplicated entries, and detailed provenance indicating source reliability.

2. Pipeline Architecture

Detect → **Extract** → **Normalize** → **CACS** → **Merge** → **Score** → **Project** → **Validate**

The deterministic, rule-based pipeline consists of advanced stages to guarantee data integrity:

1. Detect & Extract: Source-specific extractors parse inputs (CSV, JSON, GitHub, Resumes, Notes) into CandidateFragment objects.
2. Normalize: Phones map to E.164, locations extract city/country, degrees map to standard formats.
3. Conflict Analysis (CACS): A pre-merge analytical pass detects contradictory data across sources. It flags discrepancies for confidence penalties, stratifies skills by recency (Confirmed, Current, Historical), and generates a fully traceable Lineage IR.
4. Merge: Identities match via primary keys. Scalar fields take the most reliable source's value. Array fields merge via rapidfuzz deduplication.
5. Confidence Scoring: Profiles receive a score based on source weights and cross-source agreement, adjusted by CACS penalties.
6. Project & Validate: Runtime JSON configurations dictate the final structure and missing-value policies before rigid schema validation.

3. Resolution Heuristics

- Identity Merging: Uses a multi-pass approach matching overlapping emails, then phone/name heuristics.
- Experience/Education Dedup: Employs rapidfuzz matching (threshold > 75%) on institution and company names, along with date range intersections.
- Unstructured Parsing: Regex-driven contextual extraction handles phrases like 'works as a PM at Flipkart' to properly anchor both title and company, minimizing hallucinations.

4. User Interfaces

- Command Line Interface (CLI): The primary robust engine for batch processing profiles.
- Web UI (Streamlit): An interactive dashboard for evaluators to upload candidate files and view the resulting canonical JSON instantaneously.